

CYBOX // Pentesting & eJPTv2 Master Operator Guide

CYBOX is a tactical offensive state machine and interactive methodology copilot designed for penetration testers, CTF competitors, and certification candidates (**eJPTv2**, **eCPPT**, **OSCP**, **CRTP**, **PNPT**).

It replaces scattered notes with a structured, keyboard-driven terminal dashboard that tracks attack surface coverage, identifies next high-probability attack vectors, prevents rabbit-holes, auto-extracts proof flags, and exports clean executive reports.

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1. Quick-Start Cheatsheet & Keyboard Shortcuts

CYBOX is designed for pure keyboard speed without leaving the terminal.

Key / Hotkey	Action	Purpose / Context
<code>Ctrl + K</code>	Command Palette	Search and trigger any action, tab, or tool without memorizing keys
<code>Ctrl + P</code>	Fuzzy Jump	Instantly search and jump to any Target, Port, Service, Cred, or Lead

Key / Hotkey	Action	Purpose / Context
Ctrl + L	Workspace / Lab Switcher	Hot-swap lab databases, create new labs, or clone attempts
. (period)	Workspace Scratchpad	Instant freeform markdown scratchpad for raw notes & observation dumps
1 – 5	Tab Switcher	1 Workbench • 2 Creds • 3 Leads • 4 Evidence • 5 Pivots
a	Add Target	Add individual IP(s), hostnames, or entire CIDR subnets (10.10.11.0/24)
i	Initial Recon	Launch Phase-0 Recon on selected target (Nmap, Rustscan, Masscan)
r	Run Recipe	Open the selected methodology command in the interactive runner dialog
n	Triage (Next Move)	Computes the highest-probability attack vector from active state
s	I'm Stuck	Analyzes dead ends, un-sprayed credentials, and unexplored attack surface
Space	Cycle Status	Toggles check status (TODO → RUNNING → CHECKED → FINDING → DEAD_END)
t	Cycle Credential	On Creds tab: cycles test status (untested → valid [green] → invalid [red])
o	Toggle Scope	Mark selected target In-Scope or Out-of-Scope (OOS targets filtered from next moves)
c	Add Credential	Save discovered password, NTLM hash, Kerberos ticket, or SSH key to vault
l	Add Lead	Log attack hypothesis (SQLi, SMB relay, privilege escalation path)
e	Add Evidence / Flag	Capture proof flag with automatic OffSec / CTF validation
p	Methodology Profile	Switch methodology framework (eJPTv2 , Network Pentest , Web App)
g	Guided Workflow	View phase-by-phase completion progress for the active target
x	Export Report	Export engagement to Notion Workspace, Obsidian Vault, Markdown, or JSON
T	Theme Switcher	Switch color palette (Claudish, Claudish Light, Cyb0x dark)
? / F1	Interactive Help	View on-screen shortcut reference guide
q	Quit	Exit the TUI (all state is persistently saved in SQLite)

2. Environment Setup & Diagnostics

Before beginning an exam or lab engagement, run `cyb0x doctor` from your terminal to verify that your Python runtime, SQLite engine, and external penetration testing tools are ready.

```
# Verify environment health and security tooling
cyb0x doctor
```

CYBOX checks for 20+ essential security tools across five core categories:

- Port Scanning & Recon: `nmap`, `rustscan`, `masscan`
- Web Enumeration: `ffuf`, `gobuster`, `feroxbuster`, `nikto`, `whatweb`, `sqlmap`, `curl`
- Active Directory & SMB: `netexec` (`nxc`), `crackmapexec`, `smbclient`, `enum4linux-ng`, `rpcclient`
- Password Cracking & Auth: `hydra`, `john`, `hashcat`
- Network Pivoting & Tunnels: `ligolo-ng`, `chisel`, `proxychains4`

3. Step-by-Step eJPTv2 & Lab Attack Workflow

```
flowchart LR
    A[Phase 0: Setup Workspace] --> B[Phase 1: Ingest Scans / Recon]
    B --> C[Phase 2: Target 360° Assessment]
    C --> D[Phase 3: Execute Methodology Recipes]
    D --> E{Stuck or Flag Hit?}
    E -- Need Guidance --> F[Phase 4: Triage 'n' / Stuck 's']
    E -- Creds Found --> G[Phase 5: Spray & Test Creds 't']
    E -- Lateral Movement --> H[Phase 6: Pivoting Sentinel]
    F --> D
    G --> D
    H --> D
    D --> I[Phase 7: Export Report 'x']
```

Phase 0: Workspace & Scope Initialization

1. Launch a dedicated workspace for your lab or exam:

```
bash cyb0x -w ejpt-exam tui
```

(Or press `Ctrl + L` inside the TUI to create or switch workspaces on the fly).

2. Add target hosts or subnets:

3. Press `a` in the TUI.

4. Enter your target IP (e.g. `10.10.11.200`) or subnet (`192.168.1.0/24`).

5. If you already know specific open ports from your exam letter, enter them (e.g. `22,80,445,3389`).
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Phase 1: Host Discovery & Scan Ingestion

You can discover services in two ways:

Method A: Direct Ingestion from CLI

Run your scanner tools and pipe or save output directly to Cyb0x:

```
# Nmap XML or Text scan ingestion
nmap -sV -sC -p- 10.10.11.200 -oX scan_monolith.xml
cyb0x -w ejpt-exam ingest scan_monolith.xml

# Fast Rustscan JSON ingestion
rustscan -a 10.10.11.200 -g -oJ rustscan.json
cyb0x -w ejpt-exam ingest rustscan.json

# NetExec / SMB discovery output
nxc smb 10.10.11.0/24 | tee netexec_smb.txt
cyb0x -w ejpt-exam ingest netexec_smb.txt
```

Method B: In-TUI Initial Reconnaissance

1. Select the target host in the left sidebar tree.
2. Press **i** (or **r** on an unscanned host).
3. Choose a recon recipe:
4. **Fast Port Discovery (Rustscan/Nmap top-1000)**
5. **Comprehensive Service Fingerprinting (Nmap -sV -sC -p-)**
6. **Stealth SYN Scan**
7. Press **Ctrl + R** in the runner modal. Cyb0x executes the scan in the background and **automatically attaches all discovered ports, service versions, banners, and methodology checklists directly to the target tree.**

Phase 2: Target 360° Assessment & Service Enumeration

When clicking any target in the sidebar, CYBOX displays the **Target 360° Unified Operational Card**:

- **Status & Scope:** Live badge (**[DISCOVERED]** , **[FOOTHOLD]** , **[PWNERD]**) and in-scope verification.
- **Operating System Detection:** Fingerprinted OS details (e.g. **Linux (Ubuntu 22.04 LTS)** , **Windows Server 2022**).
- **Attack Surface Summary:** Total services, methodology completion %, and confirmed valid credentials.
- **Discovered Ports Table:** Complete table of all open ports, protocols, identified software versions, and check coverage.

Phase 3: Interactive Methodology Execution & Flag Capture

1. **Select a service** (e.g. `80/tcp http Apache httpd 2.4.49` or `445/tcp smb`).
 2. Cyb0x automatically populates standard eJPTv2 / OSCP methodology action items:
 3. *HTTP Directory & VHost Bruteforcing* (`ffuf` , `gobuster`)
 4. *Known CVE Checks* (e.g. *Path Traversal*, *Log4j*, *RCE exploits*)
 5. *SMB Anonymous Share Listing* (`netexec` , `smbclient`)
 6. *Active Directory Kerberoasting & AS-REP Roasting*
 7. **Execute customized commands:**
 8. Highlight any check item and press `r` .
 9. The command is pre-filled with target IP, port, and wordlists (fully editable).
 10. Press `Ctrl + R` to execute asynchronously.
 11. **Automatic Flag Detection (`▸ FLAG HIT`):**
 12. When command output contains proof flag patterns (`HTB{...}` , `flag{...}` , `user.txt` , `root.txt`),
Cyb0x's real-time regex engine **instantly flashes a prominent Flag Hit alert**.
 13. Press `Ctrl + S` to save the output and proof directly into the Evidence Ledger.
 14. **Cycle check status:**
 15. Press `Space` to cycle status: `CHECKED` (completed benign), `FINDING` (vulnerability identified), `DEAD_END` (inconclusive).
-

Phase 4: State-Aware Triage (`n`) & Rabbit-Hole Detection (`s`)

Exam success depends on avoiding 2-hour rabbit holes. CYBOX features a built-in deterministic assessment engine:

1. State-Aware Triage (`n`)

Press `n` at any time to open the **Triage Assessment Window**:

- **Known Surface vs Unexplored**: Break down what has been enumerated versus remaining blind spots.
- **Top Next Action Recommendation**: Cyb0x evaluates the exact matrix of open ports, untested services, unsprayed credentials, and priority leads to suggest the single highest-value next step.

2. Rabbit-Hole & "I'm Stuck" Escape Analysis (`s`)

Press `s` when you feel blocked:

- Detects if you have marked multiple items as dead-ends on one service while leaving high-value services (SMB, Web, RPC) completely untested.
 - Checks if you have harvested credentials that have not yet been tested on sister machines in the lab subnet.
 - Highlights unexploited high-severity leads.
-

Phase 5: Credential Vault Management & Password Spraying

Switch to **Tab 2: Credential Vault Matrix** (`2`):

1. **Record new credentials** (**c**):
 2. Enter username, secret (password, NTLM hash, SSH private key), domain (**MONOLITH.LOCAL**), and target.
 3. **Track validation across multiple hosts** (**t**):
 4. Highlight a credential and press **t** to cycle its validity on the target:
 - **UNTESTED** (gray) → **VALID ACCESS** (green) → **INVALID** (red)
 5. **Password Spraying**:
 6. As you pivot through a subnet, Cyb0x tracks which credentials have been confirmed against which machines, preventing redundant lockouts or duplicate sprays.
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Phase 6: Hypotheses, Leads & Scratchpad (**.**)

Attack Leads Board (**tab-Leads** / **3**)

- Press **l** to add hypotheses (e.g. "Check sudo -l on www-data for NOPASSWD vim", "Potential AS-REP Roasting on svc_sql").
- Organize leads with severity ratings (**CRITICAL** , **HIGH** , **MEDIUM** , **LOW**) and status (**BACKLOG** → **IN_PROGRESS** → **CONFIRMED** → **REJECTED**).

Persistent Scratchpad (**.**)

- Press **.** from anywhere in the TUI to open the **Workspace Scratchpad**.
 - Write freeform markdown notes, copy-paste exploit snippets, or jot down exam observations.
 - Press **Ctrl + S** or **ESC** — notes are immediately persisted in SQLite metadata.
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Phase 7: Network Pivoting & Route Sentinel

Switch to **Tab 5: Pivoting & Route Sentinel** (**5**):

- Document dual-homed compromised hosts acting as jump boxes.
 - Track internal victim subnets (e.g. **172.16.1.0/24**).
 - Document active tunnel bindings:
 - *Ligolo-ng tun interfaces*
 - *Chisel SOCKS5 proxies* (**127.0.0.1:1080**)
 - *SSH dynamic port forwarding*
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Phase 8: Instant Report & Vault Generation (**x**)

When you finish the assessment or need documentation for your exam report:

1. Press **x** inside the TUI (or run **cyb0x export** from CLI).
2. Choose your export format:

- 3. **Notion Assessment Workspace:** Full database structure with targets, services, credentials, and evidence.
- 4. **Obsidian Knowledge Vault:** Interlinked `.md` notes with frontmatter tags and graph view relations.
- 5. **Markdown Engagement Report:** Clean executive summary with methodology checklist completion metrics, findings list, and proof flags.
- 6. **JSON Backup:** Full structured state snapshot for backup or grading.

4. Methodology Profiles

CYBOX methodology checklists are driven by YAML profiles located in `src/cyb0x/methodology/profiles/`:

- `ejptv2.yaml` : Aligned with eJPTv2 domains (Host & Network Penetration Testing, Web Application Penetration Testing, Assessment & Auditing).
- `network_pentest.yaml` : Focuses on network services, active directory exploitation, and internal pivoting.
- `web_application.yaml` : Focuses on OWASP Top 10, directory discovery, parameter fuzzing, and API security.

Press `p` in the TUI to switch between active methodology profiles on the fly.

5. Multi-Attempt Lab Strategies & Workspace Cloning

For certification prep and practice labs where you repeat boxes for speed:

1. Press `Ctrl + L` to open the **Workspace Manager**.
2. Select your original lab workspace (e.g. `ejpt-lab08`).
3. Click "Clone for Attempt 2" (`btn-clone`).
4. Cyb0x creates an isolated clone (`ejpt-lab08_attempt2.db`) allowing you to re-execute your attack run from scratch while keeping your previous notes and proof benchmarks safely preserved.

6. CLI Command Reference

Command Line	Purpose
<code>cyb0x tui</code>	Launch the interactive terminal user interface
<code>cyb0x -w <name> tui</code>	Launch TUI directly into a named workspace
<code>cyb0x doctor</code>	Run environment readiness and tool checks
<code>cyb0x list-workspaces</code>	List all local workspace databases, targets count, findings, and flags
<code>cyb0x list-targets</code>	Print table of all registered targets and open ports

Command Line	Purpose
<code>cyb0x list-creds</code>	Display all credentials in vault (passwords masked by default)
<code>cyb0x list-commands</code>	View persistent command execution history and execution times
<code>cyb0x next <target_ip></code>	Print AI / State-Aware copilot next move recommendations to stdout
<code>cyb0x ingest <scan_file></code>	Ingest Nmap XML/gnmap, Rustscan JSON, Masscan JSON, or NetExec text
<code>cyb0x export -f notion -o ./notion</code>	Generate Notion assessment workspace
<code>cyb0x export -f obsidian -o ./vault</code>	Generate Obsidian Markdown knowledge vault
<code>cyb0x export -f markdown -o report.md</code>	Generate single Markdown engagement report
<code>cyb0x export -f json -o backup.json</code>	Export full workspace JSON state
<code>cyb0x import-backup backup.json</code>	Restore workspace state from JSON backup

CYBOX — Built for Precision, Speed, and Methodological Rigor in Offensive Security.